

Inference

Grad admissions data

Interested in predicting grad admission from GPA, GRE score, and school rank:

$$\begin{aligned}\text{Admit}_i &\sim \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ &\quad \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i\end{aligned}$$

Suppose we want to estimate the effect of GRE, after accounting for GPA and school rank. How would I compute a confidence interval for β_2 ?

Previously: confidence intervals

Normal confidence interval for β_2 :

↑
using normal
distribution

$$\hat{\beta}_2 \pm z^* SE_{\hat{\beta}_2}$$

critical value from $N(0,1)$
e.g. $z^* = 1.96$ for 95%
 $z^* = 1.645$ for 90%

Options for estimating $SE_{\hat{\beta}_2}$:

- + Bootstrap estimate of standard error
- + Estimate of standard error from R output

Confidence interval

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.989979073	1.139950936	-3.500132	0.0004650273
gpa	0.804037549	0.331819298	2.423119	0.0153878974
gre	0.002264426	0.001093998	2.069864	0.0384651284
as.factor(rank)2	-0.675442928	0.316489661	-2.134171	0.0328288188
as.factor(rank)3	-1.340203916	0.345306418	-3.881202	0.0001039415
as.factor(rank)4	-1.551463677	0.417831633	-3.713131	0.0002047107

95% CI for β_2 : $0.00226 \pm 1.96 (0.00109)$
 $= (0.00012, 0.0044)$

95% CI for e^{β_2} : $(e^{0.00012}, e^{0.0044}) = (1.00012, 1.0044)$

How do I interpret this confidence interval?

We are 95% confident that a one-point increase in GRE score is associated with an increase in odds of acceptance by between 0.012% and 0.44%, holding GPA & school rank fixed

Another question

$$\begin{aligned}\text{Admit}_i &\sim \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ &\quad \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i\end{aligned}$$

Research question: Is there a relationship between GRE and grad school acceptance, after accounting for GPA and school rank?

How could we address this research question?

Hypothesis testing

What steps are involved in hypothesis testing?

- choose desired type I error rate α
- specify H_0 & H_A
- compute a test statistic
- P-value : how unusual is the observed test statistic, if H_0 is true?
- Null distribution : distribution of test stat. under H_0

Specifying the hypotheses

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i$$

Research question: Is there a relationship between GRE and grad school acceptance, after accounting for GPA and school rank?

$$H_0: \beta_2 = 0 \quad (\text{no relationship})$$

$$H_A: \beta_2 \neq 0 \quad (\text{some relationship})$$

Confidence intervals and hypothesis testing

Research question: Is there a relationship between GRE and grad school acceptance, after accounting for GPA and school rank?

$$H_0 : \beta_2 = 0 \quad H_A : \beta_2 \neq 0$$

Recall that we would **reject** H_0 at significance level α if and only if a $(1 - \alpha)100\%$ confidence interval for β_2 **does not** contain 0. (Equivalently, a $(1 - \alpha)100$ confidence interval for e^{β_2} **does not** contain 1).

$$95\% \text{ CI for } \beta_2 : (0.00012, 0.0044)$$

$$95\% \text{ CI for } e^{\beta_2} : (1.00012, 1.0044)$$

$$\underline{\text{reject}} H_0 : \beta_2 = 0 \quad \text{at level } \alpha = 0.05$$

Computing a p-value

linear: $y = \beta_0 + \beta_1 x + \varepsilon$
 $\varepsilon \sim N(0, \sigma^2)$
 $\uparrow$
need to estimate!

To compute a p-value, we need:

logistic regression: no σ^2 to estimate

- + a test statistic
- + the distribution of the test statistic if H_0 were correct

What test statistic did we use for testing single coefficients in *linear* models?

$$H_0: \beta_j = b$$

statistic:
$$\frac{\hat{\beta}_j - b}{SE \hat{\beta}_j}$$

same idea for logistic regression!

$$H_0: \beta_2 = 0 \quad Z = \frac{\hat{\beta}_2 - 0}{SE \hat{\beta}_2} = 2.07$$

The null distribution

To compute a p-value, we need the **null distribution** (distribution of the test statistic under H_0).

Does *bootstrap resampling* (what we used to create a confidence interval) give us the **null** distribution?

Not quite !

Bootstrap resampling mimics sampling
from population
we don't know if H_0 is true in
population !

Parametric bootstrapping

if H_0 is true: don't need GRE!

→ Reduced model: $\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 RankZ_i + \beta_3 RankB_i + \beta_4 RankU_i$

① Fit reduced model

② Simulate response from reduced model (H_0 guaranteed true!) $Admit_i^* \sim \text{Bernoulli}(\hat{\pi}_{i, \text{reduced}})$

③ Refit full model (GRE included!) on simulated data → $\hat{\beta}_{\text{sim}}$

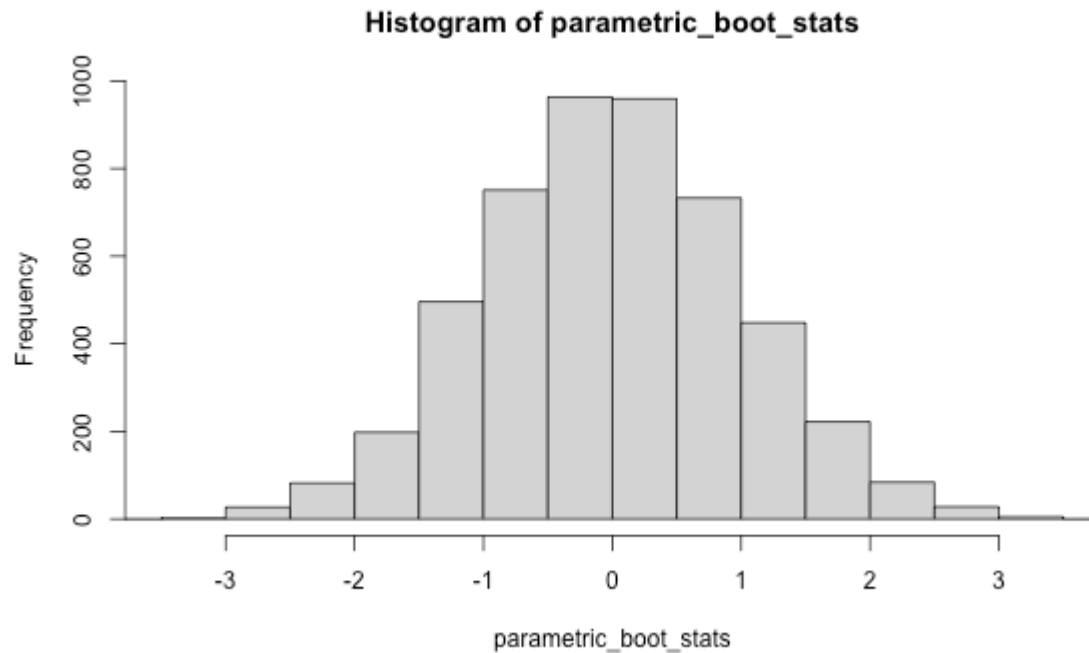
④ Recalculate test statistic: $Z_{\text{sim}} = \frac{\hat{\beta}_{\text{sim}}}{SE \hat{\beta}_{\text{sim}}}$

⑤ Repeat many times!

Class activity

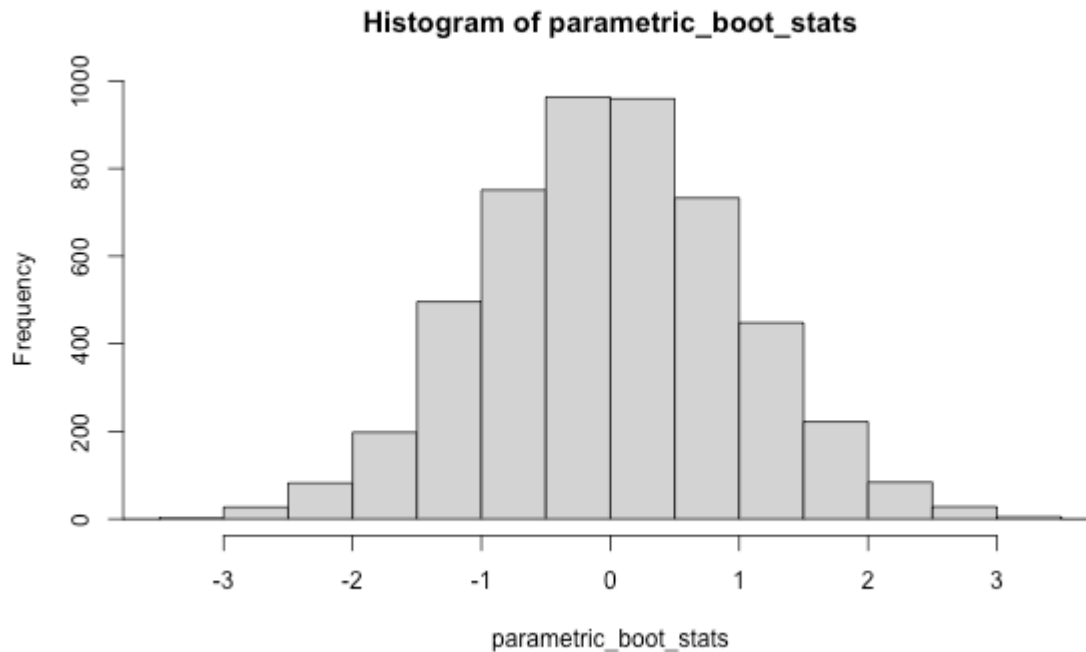
- + Work on the class activity (linked on Canvas and course website)
- + Submit rendered HTML on Canvas at the end of class

Activity



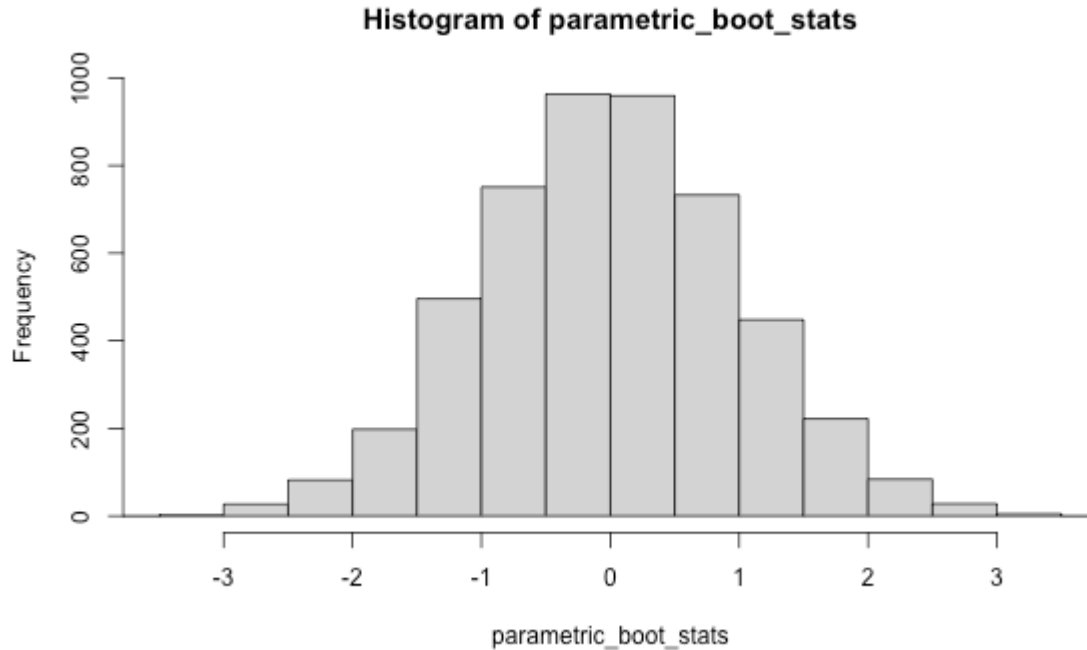
What does this distribution look like?

Activity



How unusual is the observed test statistic of 2.07, compared to this null distribution?

p-value



```
mean(abs(parametric_boot_stats) > 2.07)
```

```
## [1] 0.0394
```

p-value

```
mean(abs(parametric_boot_stats) > 2.07)
```

```
## [1] 0.0394
```

```
summary(m_full)$coefficients
```

##		Estimate	Std. Error	z value	Pr(> z)
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